

Gerald Truong

Senior Software Engineer |
AI-Enabled Applications |
Full-Stack & Distributed Systems

Austin, TX
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Profiles

linkedin.com/in/geraldtruong

Skills

AI & Intelligent Automation

Advanced
Agentic AI Workflows, AI-Assisted Data Mapping, LLM Integration, Automation Solutions

Backend Development & APIs

Advanced
TypeScript, Node.js, NestJS, REST APIs, GraphQL, SSO, SQL, Ruby on Rails, PHP, Python

Frontend & UI/UX Engineering

Advanced
JavaScript (ES6+), React.js, Redux, Vue.js, Angular, HTML5, CSS3, Storybook, Design Systems

Data Engineering & Visualization

Advanced
D3.js, Visualize.js, Big Data UI Components, ETL Pipelines, SVG, Data Pipeline Monitoring

EdTech & Standards

Advanced
SIS/LMS Integrations, LTI 1.1 / 1.3, Rostering APIs, Canvas, Moodle, Brightspace, Blackboard

DevOps, Testing & Tools

Intermediate
Git, Docker, Puppeteer JS (E2E Testing), Webpack, Salesforce Platform (Apex), Pendo Analytics, JIRA

Hardware & Systems Engineering

Advanced
Signal Integrity, High-Speed Digital Design, Circuit Analysis, IBIS Modeling, CAD Application Development

Senior Software Engineer with 15+ years of experience designing and delivering scalable enterprise software, APIs, data-intensive systems, and modern web applications across healthcare, education technology, cybersecurity, and business intelligence. Recently focused on building AI-enabled applications that leverage agentic workflows and intelligent automation to solve complex enterprise challenges, including AI-assisted data mapping and customer integration solutions. Proven track record of leading technical initiatives, architecting production-ready software, mentoring engineers, and delivering customer-focused solutions. Backed by a Master's degree in Electrical Engineering and a decade of experience in high-speed digital design and circuit analysis, bringing a systems-thinking approach to designing reliable, scalable software and AI solutions.

Experience

Edlink

Software Engineer (full-time)
<https://ed.link>

June 2023 to November 2025

Austin, TX

- Architected and deployed an AI-assisted data mapping agent leveraging agentic workflows to help clients map SIS fields to Edlink standards.
- Engineered rostering, SSO, and coursework API integrations for Facts SIS, Blackbaud SIS, Toddle LMS, and Moodle LMS using Node.js, NestJS, and TypeScript.
- Developed LTI 1.1 and LTI 1.3 solutions across major LMS platforms including Canvas, Schoology, Brightspace, and Blackboard.
- Served as Technical Lead for the client success team, delivering custom technical solutions to resolve complex integration challenges.
- Built data pipeline monitoring UIs, custom JSON property editors for rostering records, and SQL/Node.js data audit notification services.

Medidata

Senior Software Engineer (full-time)
<https://www.medidata.com/en/>

2017 to March 2023

Austin, TX

- Led a team of two data scientists to develop an OCR and ETL-driven budget forecasting data pipeline for clinical trial management.
- Architected core components for 1XDS, Medidata's enterprise React design system and Storybook component library.
- Designed and built template-based, quick-entry budgeting applications and complex clinical trial user workflows using React and Redux.
- Pioneered end-to-end testing frameworks across multiple core applications using Puppeteer JS.
- Integrated Pendo Analytics live chat features to enhance customer support and mentored junior developers in modern React/Redux practices.

Synack

Senior Software Engineer (full-time)
<https://synack.com>

2015 to 2017

Austin, TX

- Engineered high-performance data visualization components for Synack's security dashboard product using React, D3.js, SVG, and REST APIs.
- Developed the core Synack platform messaging application utilizing React, Redux, and Ruby on Rails.
- Translated complex security dashboard design mockups into fully responsive, production-ready web applications.

Jobscience

Technical Lead (full-time)

<https://www.linkedin.com/company/jobscience>

2013 to 2015

Austin, TX

- Developed mobile job board applications, candidate resume evaluator tools, and custom PDF viewing renderers on the Salesforce platform using Apex, JavaScript, HTML, and CSS.
- Provided technical leadership and code reviews for an offshore engineering team based in Belarus.

Jaspersoft

Senior Software Engineer (full-time)

<https://www.jaspersoft.com>

2011 to 2013

Austin, TX

- Created and developed Visualize.js, an award-winning JavaScript framework for embedding interactive dashboards and reports into external web applications.
- Developed big data UI table components capable of rendering over 100,000 tabular data rows efficiently in the browser.
- Engineered cross-browser charting components and reduced overall CSS payload sizes by 25% across Jaspersoft Server web apps.

Isola

Software Contractor (contract)

<https://www.isola-group.com/>

2010 to 2011

Austin, TX

- Architected IsoDesign, the industry's first Web CAD application for high-speed PCB stackup design using JavaScript, SVG, Raphael.js, and MVC design principles.

Force 10 Networks

Signal Integrity Engineer (full-time)

<https://en.wikipedia.org/wiki/Force10>

2006 to 2007

Austin, TX

- Executed signal integrity simulations and circuit analysis for 24-layer linecards featuring multi-gigahertz ASICs, DDR2+, QDR2+, and high-speed SerDes channels.
- Procured, validated, and maintained complex IBIS models for high-speed digital board design verification.

Northrop Grumman

Electrical Engineer / Application Support Engineer (full-time)

<https://www.northropgrumman.com/>

2003 to 2005

Austin, TX

- Conducted circuit analysis and signal integrity simulations on critical digital modules for the F-22 software radio system and satellite C&DH; subsystems.
- Created and led training programs for Mentor Graphics signal integrity simulation software serving 40+ hardware engineers.

Motorola Computer Group

Signal Integrity Engineer (full-time)

<https://emuseum.tempe.gov/objects/2767/motorola--2900-s-diablo-way--tempe-arizona>

2000 to 2002

Tempe, AZ

- Performed signal integrity analysis and IBIS model validations using Mentor Graphics and Cadence SI suites to guarantee signal quality on complex PCB layouts.
- Collaborated directly with IC vendors to acquire and review high-speed component simulation models.

Education

The University of Texas at Austin (McCombs School of Business)

Post Graduate Program in AI and Machine Learning

January 2024

Austin, TX

California State University, Long Beach

Master's in Electrical Engineering

Long Beach, CA